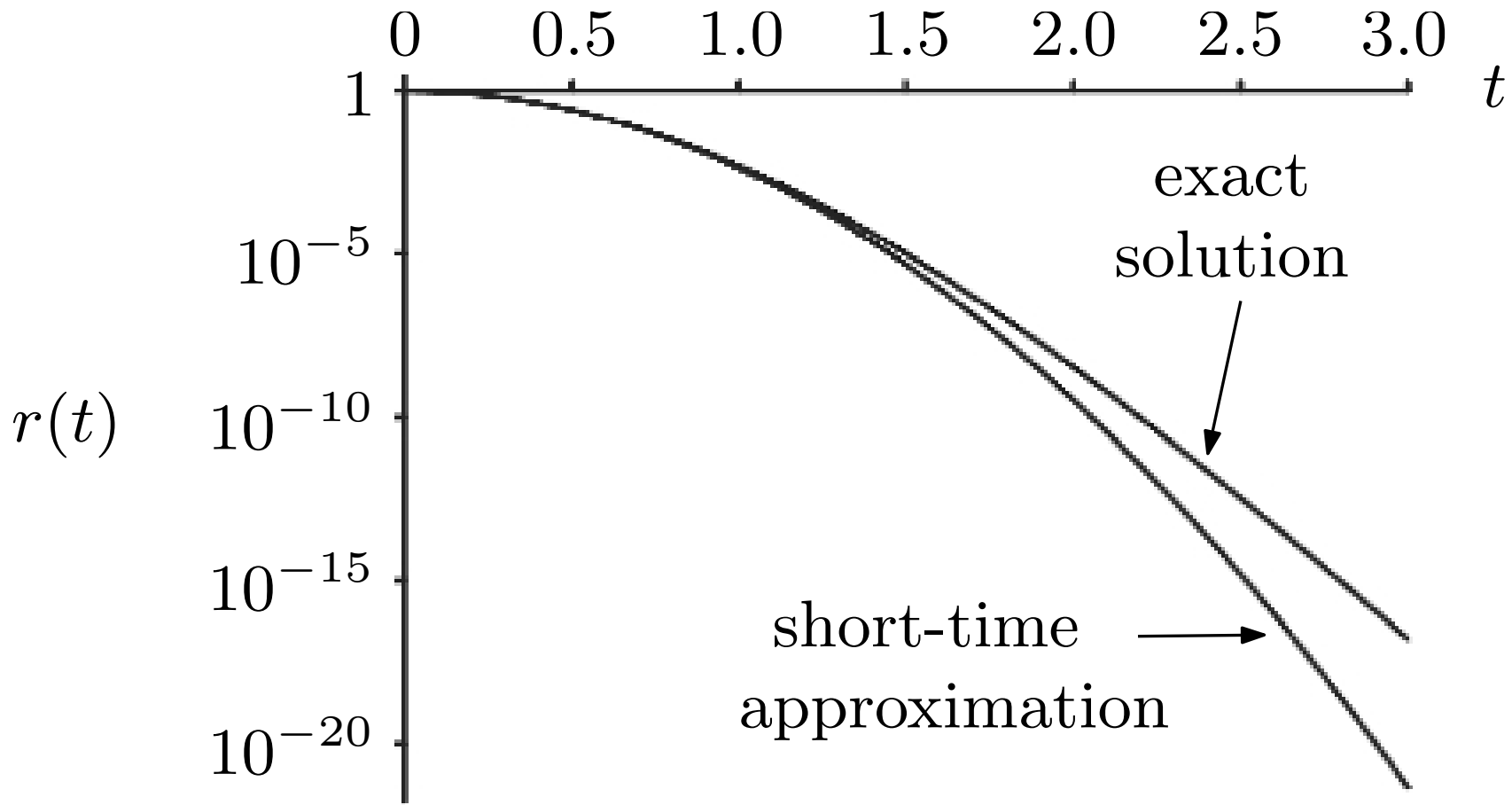

0
0.5
1.0
1.5
2.0
2.5
3.0
t
1
10^{-5}
10^{-10}
10^{-15}
10^{-20}
r(t)
exact
solution
short-time
approximation